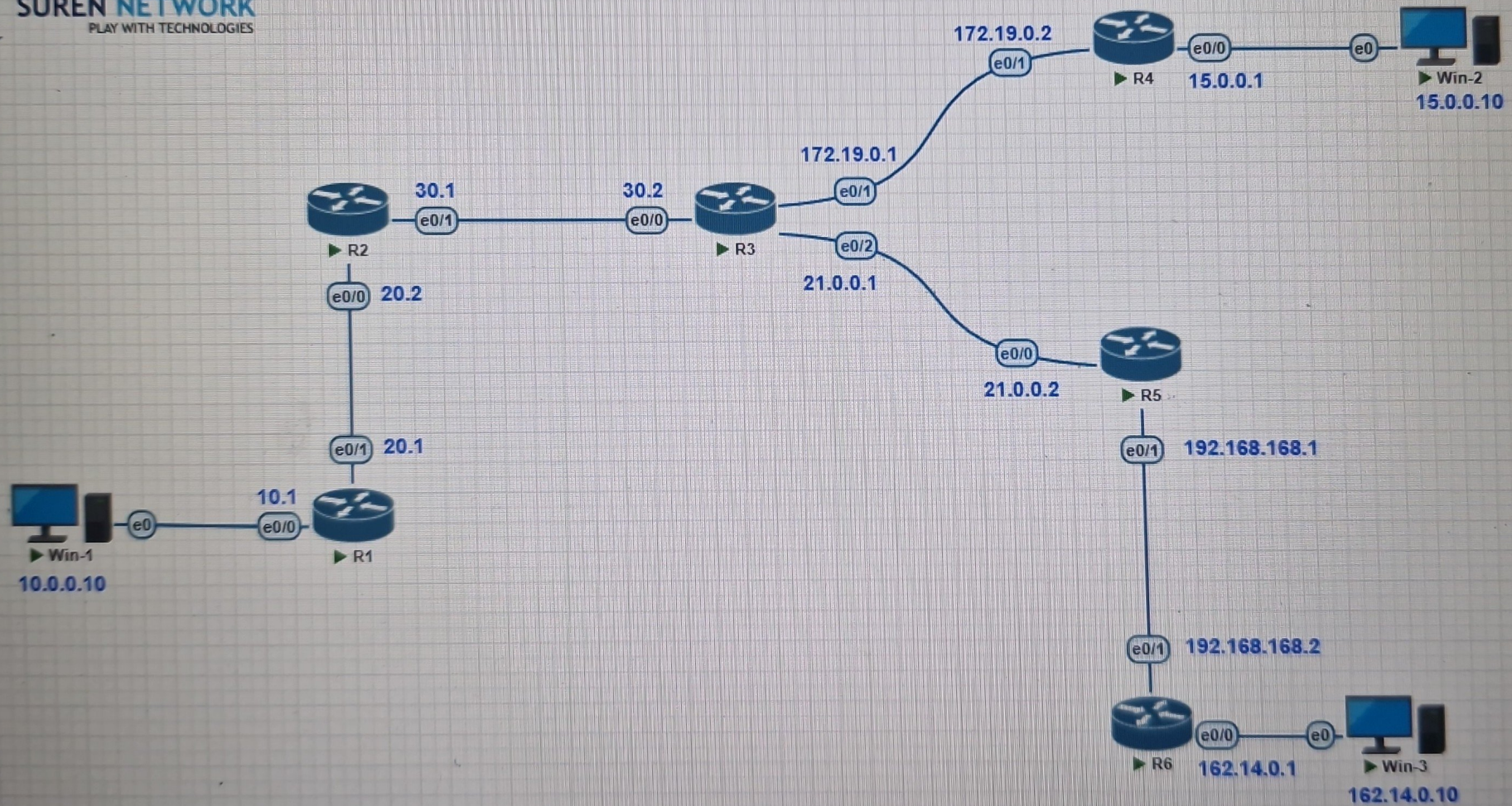




E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
 10.0.0.0/8 is directly connected, Ethernet0/0
 10.0.0.1/32 is directly connected, Ethernet0/0
15.0.0.0/8 [1/0] via 20.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
 20.0.0.0/8 is directly connected, Ethernet0/1
 20.0.0.1/32 is directly connected, Ethernet0/1
21.0.0.0/8 [1/0] via 20.0.0.2
30.0.0.0/8 [1/0] via 20.0.0.2
162.14.0.0/16 [1/0] via 20.0.0.2
172.19.0.0/16 [1/0] via 20.0.0.2
192.168.168.0/24 [1/0] via 20.0.0.2

R1#

R1#

R1#

R2

E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 20.0.0.1
S    15.0.0.0/8 [1/0] via 30.0.0.2
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        20.0.0.0/8 is directly connected, Ethernet0/0
L        20.0.0.2/32 is directly connected, Ethernet0/0
S    21.0.0.0/8 [1/0] via 30.0.0.2
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/8 is directly connected, Ethernet0/1
L        30.0.0.1/32 is directly connected, Ethernet0/1
S    162.14.0.0/16 [1/0] via 30.0.0.2
S    172.0.0.0/16 [1/0] via 30.0.0.2
S    192.168.168.0/24 [1/0] via 30.0.0.2
```

R2#

R2#

R2#

E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

I

```
S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      21.0.0.0/8 is directly connected, Ethernet0/2
L      21.0.0.1/32 is directly connected, Ethernet0/2
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      30.0.0.0/8 is directly connected, Ethernet0/0
L      30.0.0.2/32 is directly connected, Ethernet0/0
S    162.14.0.0/16 [1/0] via 21.0.0.2
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      172.19.0.0/16 is directly connected, Ethernet0/1
L      172.19.0.1/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 21.0.0.2
```

R3#

R4

E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 172.19.0.1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    15.0.0.0/8 is directly connected, Ethernet0/0
L    15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
L    172.19.0.2/32 is directly connected, Ethernet0/1
S    192.0.0.0/24 [1/0] via 172.19.0.1
R4#
R4#
R4#
```

R4 15.0.0.0

R5 192.168.0.0

R5 192.168.0.0

R6 162.14.0.0

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R5

172.19.0.2

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S 10.0.0.0/8 [1/0] via 21.0.0.1
S 15.0.0.0/8 [1/0] via 21.0.0.1
S 20.0.0.0/8 [1/0] via 21.0.0.1
C 21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 21.0.0.0/8 is directly connected, Ethernet0/0
L 21.0.0.2/32 is directly connected, Ethernet0/0
S 30.0.0.0/8 [1/0] via 21.0.0.1
S 162.14.0.0/16 [1/0] via 192.168.168.2
S 172.0.0.0/16 [1/0] via 21.0.0.1
C 192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
L 192.168.168.0/24 is directly connected, Ethernet0/1
L 192.168.168.1/32 is directly connected, Ethernet0/1
R5#
R5#
```


R6

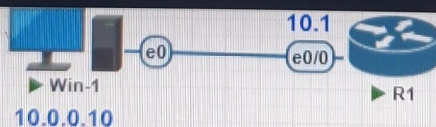
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S 10.0.0.0/8 [1/0] via 192.168.168.1
S 15.0.0.0/8 [1/0] via 192.168.168.1
S 20.0.0.0/8 [1/0] via 192.168.168.1
S 21.0.0.0/8 [1/0] via 192.168.168.1
S 30.0.0.0/8 [1/0] via 192.168.168.1
162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 162.14.0.0/16 is directly connected, Ethernet0/0
L 162.14.0.1/32 is directly connected, Ethernet0/0
S 172.19.0.0/16 [1/0] via 192.168.168.1
192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.168.0/24 is directly connected, Ethernet0/1
L 192.168.168.2/32 is directly connected, Ethernet0/1
```

R6#

R6#



QEMU (Win-1)

C:\WINDOWS\system32\cmd.exe

```
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
```

```
Ping statistics for 15.0.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

```
C:\Documents and Settings\user>ping 162.14.0.10
```

```
Pinging 162.14.0.10 with 32 bytes of data:
```

```
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
```

```
Ping statistics for 162.14.0.10:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

```
C:\Documents and Settings\user>ipconfig
```

```
Windows IP Configuration
```

```
Ethernet adapter Local Area Connection:
```

```
Connection-specific DNS Suffix . : 
IP Address . . . . . : 10.0.0.10
Subnet Mask . . . . . : 255.0.0.0
Default Gateway . . . . . : 10.0.0.1
```

```
C:\Documents and Settings\user>_
```

Start

C:\WINDOWS\system...

QEMU (Win-2)



C:\WINDOWS\system32\cmd.exe

```
C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=1ms TTL=124

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . : 
    IP Address. . . . . : 15.0.0.10
    Subnet Mask . . . . . : 255.0.0.0
    Default Gateway . . . . . : 15.0.0.1

C:\Documents and Settings\user>
```


QEMU (Win-3)

C:\WINDOWS\system32\cmd.exe

Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123

Ping statistics for 10.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=4ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124

Ping statistics for 15.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix	:	:
IP Address	:	162.14.0.10
Subnet Mask	:	255.255.0.0
Default Gateway	:	162.14.0.1

C:\Documents and Settings\user>

Start

C:\WINDOWS\system...